INCH-POUND

AN848 Rev 12
7 June 2011
SUPERSEDING
AN848 Rev 11
20 March 1979

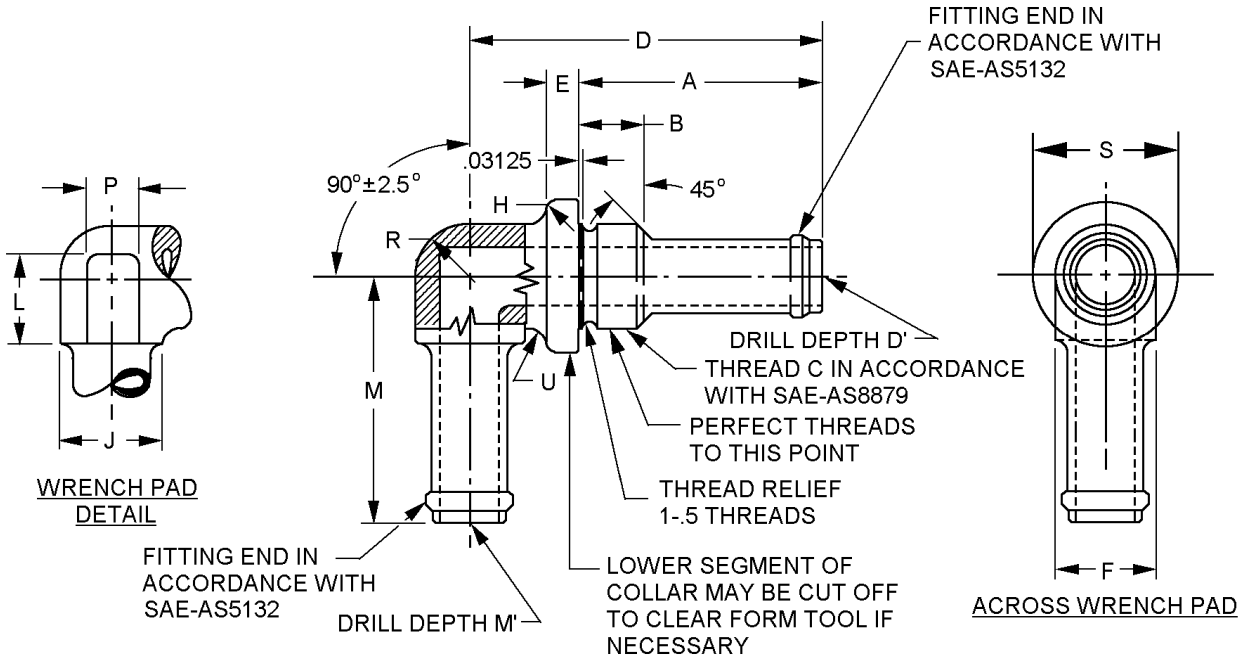
DETAIL SPECIFICATION SHEET

ELBOW, HOSE, 90°

Reinstated after 7 June 2011. Inactive for new design.
For new design, use SAE-AS5188.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and SAE-AS4843.




| Inch | mm |
|---|---|
| .01325 | 0.80 |

FIGURE 1. 90° elbow dimensions and configuration.

AMSC N/A

FSC 4730

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| Dash number | Hose ID inches (mm) | A inches (mm) | B inches (mm) | Thread C Ref SAE-AS8879 | D +.047 (1.19) -.000 inches (mm) |
|---|---|---|---|---|---|
| 4 | .250 (6.35) | 1.938 (49.23) | .438 (11.13) | .4375-20UNJF-3A | 2.063 (52.40) |
| 6 | .375 (9.53) | 1.969 (50.01) | .469 (11.91) | .5625-18UNJF-3A | 2.625 (66.68) |
| 8 | .500 (12.70) | 2.000 (50.80) | .500 (12.70) | .7500-16UNJF-3A | 2.781 (70.64) |
| 10 | .675 (17.15) | 2.031 (51.59) | .531 (13.49) | 1.875 -12UNJ-3A | 2.969 (75.41) |
| 12 | .750 (19.05) | 2.167 (55.04) | .563 (14.30) | 1.0625 -12UNJ-3A | 3.094 (78.59) |
| 16 | 1.000 (25.40) | 2.167 (55.04) | .563 (14.30) | 1.3125-12UNJ-3A | 3.219 (81.76) |
| 20 | 1.250 (31.75) | 2.167 (55.04) | .563 (14.30) | 1.6250-12UNJ-3A | 3.531 (89.69) |
| 24 | 1.500 (38.10) | 2.167 (55.04) | .563 (14.30) | 1.8750-12UNJ-3A | 3.656 (92.86) |

| Dash number | D' +.047 (1.19) -.000 inches (mm) | E inches (mm) | F inches (mm) | H Radius inches (mm) | J Diameter inches (mm) | L Approx. inches (mm) |
|---|---|---|---|---|---|---|
| 4 | 2.500 (63.50) | .156 (3.96) | .438 (11.13) | .063 (1.60) | .438 (11.13) | .313 (7.95) |
| 6 | 2.672 (67.87) | .188 (4.78) | .563 (14.30) | .063 (1.60) | .563 (14.30) | .500 (12.70) |
| 8 | 2.844 (72.24) | .219 (5.56) | .750 (19.05) | .094 (2.39) | .750 (19.05) | .625 (15.88) |
| 10 | 3.094 (78.59) | .250 (6.35) | .875 (22.23) | .094 (2.39) | .875 (22.23) | .750 (19.05) |
| 12 | 3.219 (81.76) | .250 (6.35) | 1.063 (27.00) | .094 (2.39) | 1.063 (27.00) | .938 (23.83) |
| 16 | 3.344 (84.94) | .250 (6.35) | 1.313 (33.35) | .125 (3.18) | 1.313 (33.35) | 1.000 (25.40) |
| 20 | 3.656 (92.86) | .281 (7.14) | 1.625 (41.28) | .125 (3.18) | 1.625 (41.28) | 1.125 (28.58) |
| 24 | 3.781 (96.04) | .281 (7.14) | 1.875 (47.63) | .125 (3.18) | 1.875 (47.63) | 1.500 (38.10) |

| Dash number | M +.047 (1.19) -.000 inches (mm) | M' +.047 (1.19) -.000 inches(mm) | P Approx inches(mm) | R Radius inches (mm) | S Diameter inches (mm) | U Radius inches (mm) |
|---|---|---|---|---|---|---|
| 4 | 1.781 (45.24) | 1.828 (43.43) | .250 (6.35) | .219 (5.56) | .625 (15.88) | .063 (1.60) |
| 6 | 1.875 (47.63) | 1.923 (48.84) | .313 (7.95) | .281 (7.14) | .813 (20.65) | .094 (2.39) |
| 8 | 1.969 (50.01) | 2.031 (51.59) | .438 (11.13) | .375 (9.53) | 1.000 (25.40) | .094 (2.39) |
| 10 | 2.063 (52.40) | 2.188 (55.58) | .438 (11.13) | .438 (11.13) | 1.250 (31.75) | .125 (3.18) |
| 12 | 2.156 (54.76) | 2.281 (57.94) | .500 (12.70) | .531 (13.49) | 1.438 (36.53) | .125 (3.18) |
| 16 | 2.281 (57.94) | 2.406 (61.11) | .563 (14.30) | .656 (16.66) | 1.688 (42.88) | .125 (3.18) |
| 20 | 2.500 (63.50) | 2.625 (66.68) | .563 (14.30) | .813 (20.65) | 2.063 (52.40) | .188 (4.48) |
| 24 | 2.625 (66.68) | 2.750 (69.85) | .625 (15.88) | .938 (22.83) | 2.375 (60.33) | .188 (4.48) |

| Max weight lbs (kg) | | | |
|---|---|---|---|
| Copper | Al | CRES | Ti |
| .120 (0.05) | .040 (0.02) | .120 (0.05) | .066 (0.03) |
| .180 (0.08) | .060 (0.03) | .180 (0.08) | .100 (0.05) |
| .300 (0.14) | .100 (0.05) | .300 (0.14) | .165 (0.07) |
| .390 (0.17) | .130 (0.06) | .389 (0.18) | .215 (0.10) |
| .600 (0.27) | .200 (0.09 | .599 (0.27) | .331 (0.15) |
| .870 (0.39) | .290 (0.13) | .868 (0.39) | .479 (0.23) |
| 1.500 (0.68) | .500 (0.23) | 1.497 (0.68) | .827 (0.38) |
| 2.100 (0.95) | .700 (0.32) | 2.096 (1.32) | 1.157 (0.52) |

FIGURE 1. 90° elbow dimensions and configuration - Continued.



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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.016 inch (0.41 mm).
4. Break sharp edges .and remove all hanging burrs and slivers
5. Machined surfaces shall be finished to 125µ in Ra, forged surfaces shall be 250µ inches Ra, unless otherwise specified on the figures. Surface finish shall be in accordance with ASME B46.1.
6. For design features purposes, this standard takes precedence over documents referenced herein.
7. Referenced documents shall be of the issue in effect on date of invitation for bid.

FIGURE 1. 90° elbow dimensions and configuration - Continued.

REQUIREMENTS:

Dimensions and configuration shall be in accordance with figure 1.

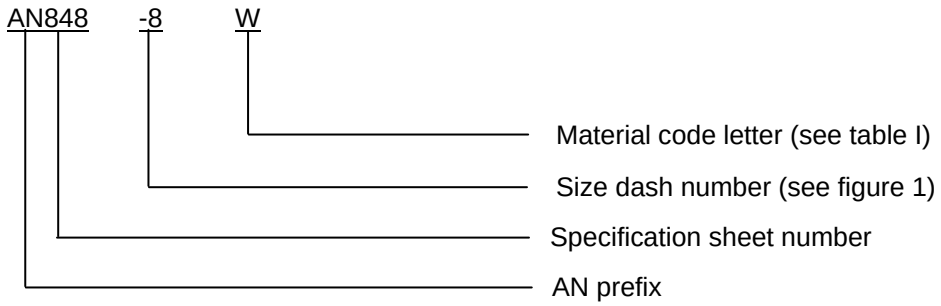
Materials and finishes shall be in accordance with SAE-AS4843, see table I for material code

TABLE I. Material and code letters.

| Code letter | Material |
|---|---|
| Blank | Copper alloy |
| J | Corrosion resistant steel (CRES), type 304 |
| K | CRES, type 316 |
| R | CRES. Type 321 |
| S | CRES, type 347 |
| T 1/ | Titanium alloy |
| W | Aluminum alloy 7075-T73 |

1/ Not for use in oxygen systems.

Part or Identifying Number (PIN): The PIN consists of the letter “AN” the specification sheet number, a dash number for tube and port size, and a material code letter for material type. Unassigned PIN’s shall not be used.




PIN example: AN848-8WP indicates a 90° elbow hose to hose, .500 inch (12.70 mm) hose, aluminum alloy 7075-T73.

Supersession data. The aluminum “D” designator has been replaced by the “W” designator.

Marking: Part shall be permanently marked with the AN PIN, and include the manufacturers CAGE, name, or trademark.



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Table II provides a detailed cross-reference of AN848 PINs and replacement SAE-AS5188 PINs. Users are cautioned to evaluate replacements for their particular application.

CAUTION: The superseding information is valid as of the date of this specification and may be superseded by subsequent revisions of the superseding document.

TABLE II. Cross-reference data. 1/

| AN PIN | Hose Size | Replacement AS PIN | New design |
|---|---|---|---|
| AN848-4 | 0.250 | AS5188B04 | |
| AN848-4D | 0.250 | AS5188W04 | AS5188W04 |
| AN848-4J | 0.250 | AS5188J04 | |
| AN848-4K | 0.250 | AS5188K04 | |
| AN848-4R | 0.250 | AS5188R04 | |
| AN848-4S | 0.250 | AS5188R04 | AS5188R04 |
| AN848-4T | 0.250 | None | |
| AN848-4W | 0.250 | AS5188W04 | |
| AN848-6 | 0.375 | AS5188B06 | |
| AN848-6D | 0.375 | AS5188W06 | AS5188W06 |
| AN848-6J | 0.375 | AS5188J06 | |
| AN848-6K | 0.375 | AS5188K06 | |
| AN848-6R | 0.375 | AS5188R06 | |
| AN848-6S | 0.375 | AS5188R06 | AS5188R06 |
| AN848-6T | 0.375 | None | |
| AN848-6W | 0.375 | AS5188W06 | |
| AN848-8 | 0.500 | AS5188B08 | |
| AN848-8D | 0.500 | AS5188W08 | AS5188W08 |
| AN848-8J | 0.500 | AS5188J08 | |
| AN848-8K | 0.500 | AS5188K08 | |
| AN848-8R | 0.500 | AS5188R08 | |
| AN848-8S | 0.500 | AS5188R08 | AS5188R08 |
| AN848-8T | 0.500 | None | |
| AN848-8W | 0.500 | AS5188W08 | |
| AN848-10 | 0.625 | AS5188B10 | |
| AN848-10D | 0.625 | AS5188W10 | AS5188W10 |
| AN848-10J | 0.625 | AS5188J10 | |
| AN848-10K | 0.625 | AS5188K10 | |
| AN848-10R | 0.625 | AS5188R10 | |
| AN848-10S | 0.625 | AS5188R10 | AS5188R10 |
| AN848-10T | 0.625 | None | |
| AN848-10W | 0.625 | AS5188W10 | |

See note at end of table.



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TABLE II. Cross-reference data - Continued. 1/

| AN PIN | Hose Size | Replacement AS PIN | New design |
|---|---|---|---|
| AN848-12 | 0.750 | AS5188B12 | |
| AN848-12D | 0.750 | AS5188W12 | AS5188W12 |
| AN848-12J | 0.750 | AS5188J12 | |
| AN848-12K | 0.750 | AS5188K12 | |
| AN848-12R | 0.750 | AS5188R12 | |
| AN848-12S | 0.750 | AS5188R12 | AS5188R12 |
| AN848-12T | 0.750 | None | |
| AN848-12W | 0.750 | AS5188W12 | |
| AN848-16 | 1.000 | AN5188B16 | |
| AN848-16D | 1.000 | AS5188W16 | AS5188W16 |
| AN848-16J | 1.000 | AS5188J16 | |
| AN848-16K | 1.000 | AS5188K16 | |
| AN848-16R | 1.000 | AS5188R16 | |
| AN848-16s | 1.000 | AS5188R16 | AS5188R16 |
| AN848-16T | 1.000 | None | |
| AN848-16W | 1.000 | AS5188W16 | |
| AN848-20 | 1.250 | AS5188B20 | |
| AN848-20D | 1.250 | AS5188W20 | AS5188W20 |
| AN848-20J | 1.250 | AS5188J20 | |
| AN848-20K | 1.250 | AS5188K20 | |
| AN848-20R | 1.250 | AS5188R20 | |
| AN848-20S | 1.250 | AS5188R20 | AS5188R20 |
| AN848-20T | 1.250 | None | |
| AN848-20W | 1.250 | AS5188W20 | |
| AN848-24 | 1.500 | AS5188B24 | |
| AN848-24D | 1.500 | AS5188W24 | AS5188W24 |
| AN848-24J | 1.500 | AS5188J24 | |
| AN848-24K | 1.500 | AS5188K24 | |
| AN848-24R | 1.500 | AS5188R24 | |
| AN848-24S | 1.500 | AS5188R24 | AS5188R24 |
| AN848-24T | 1.500 | None | |
| AN848-24W | 1.500 | AS5188W24 | |

1/ For new design use material designators R or W.

Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to SAE-AS4843, this document references the following:

ASME B46.1
SAE-AS5132
SAE-AS5188
SAE-AS8879



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## CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2011-060)

Review activity:
Air Force - 71

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